

A field-level dark-siren likelihood

Making the large-scale-structure field a latent variable of the hierarchy, and recovering catalog sirens and cross-correlation sirens as its two limits

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Executive summary

The dark-siren likelihood in this repository already contains a three-dimensional Bayesian reconstruction of the galaxy density field. It is used in the wrong place. The field is fitted once, offline, at a fixed cosmology, a fixed mean density and a fixed completeness fit; it is exported to disk as a table $Q_p(z)$; and a provenance firewall then *forbids* the inference from sampling any of the parameters the table was built at. The literal sentence “every cosmological proposal changes the galaxy-field likelihood consistently” is, today, unrepresentable by construction.

The proposal. Delete the table. Promote the whitened low-rank coefficients ξ of the field to a latent variable of the hierarchy, so that one field appears in three places at once: in the galaxy *counts*, which constrain it; in the *placement* of missing hosts, which the gravitational-wave events probe; and in the *selection normalisation*, which decides what fraction of universes like this one are detectable. Every cosmological proposal then re-weights the galaxy-field likelihood, and the missing-host field it delivers to the event terms and to the detected fraction is the same realisation.

Why this is more than plumbing. The same generative hierarchy contains the two established dark-siren analyses as limits of one likelihood. When the catalog is complete the missing branch vanishes and the field drops out *identically*: one recovers the discrete weighted-host sum. When the catalogued fraction of an event’s localisation volume tends to zero and the volume spans many correlation cells, marginalising the Gaussian latent analytically replaces the field by its kernel, and the surviving event $\times$ galaxy-voxel term is the harmonic cross-correlation $C_\ell^{\text{GW} \times g}$ carrying the bias combination $b_{\text{GW}} b_{\text{gal}}$. The interior — an incomplete catalog with well-localised events — is where every real catalog/detector pairing actually sits, and it is the regime neither limit describes. Which corner a given pairing occupies is set by two numbers one can measure, not by which method one prefers.

What is affordable, and what is not. A full per-proposal Newton re-solve of the field is not affordable: at the smallest rank the resolution guards permit ($M = M_{\text{sph}} M_z = 315 \times 12 = 3780$) it is ~ 1 s of dense linear algebra against a measured 27.5–49.3 ms likelihood, a factor of ~ 36 , inside a run that makes $\sim 10^6$ calls. The affordable form is *linear response*: the mode of the field posterior moves affinely with the proposal through $S = \partial \hat{\xi} / \partial \theta$, obtained from the implicit function theorem against the Cholesky factor the anchor solve already computes, at a storage cost of 121 kB and an online cost of one member-independent row-factor contraction. The correction is

member-independent, which is the single fact that keeps the memory budget flat under a 256-fold evaluation concurrency.

What must be conceded in the same breath. Three things, and they belong in the abstract, not the appendix. (i) The galaxy field constrains the *shape* parameters $(\Omega_m, w_0, w_a, \delta)$ and the tracer biases; it constrains H_0 only through the volume–density amplitude $n_0(c/H_0)^3$, which is exactly degenerate with the mean comoving number density — so an H_0 constraint from the counts is manufactured unless n_0 carries an H_0 -aware prior. (ii) The field is defined over the volume the survey sees: $Q \equiv 1$ off the footprint (38% of the sky) and above the survey depth $z_{\text{depth}} = 0.3$. Four independent integrals put the in-support fraction of the missing-galaxy budget at 2.2×10^{-5} to 2.7×10^{-4} . The lowest and most defensible is the verification integral for this document, which uses the shipped fit *including its cubic $K(z)$ template* on the production $z_{\text{max}}=6$ grid: 2.2×10^{-5} all-sky, 1.4×10^{-5} restricted to the occupied footprint. The K -correction is why it sits below the other three: it drives C_{sel} down faster with redshift ($C_{\text{sel}} = 0.92$ at z_{depth} , $< 10^{-3}$ by $z = 1$), moving yet more of the missing budget above the survey depth, where $Q \equiv 1$. The field is being asked to redistribute two thousandths of one percent of the missing budget. (iii) The mean-one budget constraint that makes “ C and n_0 own the budget, Q owns placement” true is a *gauge fixing*, not a measurement.

The first thing to do is the cheapest thing to do. Point (ii) is not a caveat, it is a falsifiable prediction that the whole direction may be a null operation, and it is testable in three days against artefacts that exist today: the event-weighted fraction of prior mass in the in-support missing branch, and the oscillation of the log-likelihood across the H_0 prior when the *existing* table is switched on at the anchor. That second number is an upper bound on everything the entire programme can buy. If both are small, the honest deliverable is a bounded systematic — “large-scale-structure placement uncertainty contributes less than X to the H_0 posterior” — and it is a publishable result reachable in days rather than weeks. Every other measurement in this document is downstream of that one.

What is genuinely new, stated at the level the literature permits. Three-dimensional Bayesian reconstruction of the galaxy field for dark sirens is not new. Lognormal density reconstruction is not new. Multiplicative large-scale-structure corrections to catalog completion are not new. What is not found in the completion literature is the conjunction: a *shared* latent field carrying per-tracer bias and per-tracer parametric selection, marginalised inside a finite-event gravitational-wave hierarchy with the same realisation in the event terms and in the detection normalisation, at fixed missing-budget monopole — and the demonstration that catalog sirens and cross-correlation sirens are its two limits. The claim to *first* is not made here; the paper-by-paper search that would license it is the first item of work.

1 What the machinery does today

The pieces are usually described as three machines — a builder, a table, a likelihood. They are one generative model, and writing them as one is what exposes where the model is exact and where it is a plug-in. All equations marked [CODE] are transcribed from verified source lines at `master 0c5b3db`.

1.1 Notation

Sky pixels are $p \in \{1, \dots, N_{\text{pix}}\}$ in HEALPix RING ordering at $N_{\text{side}} = 64$ ($N_{\text{pix}} = 49\,152$, $\Omega_{\text{pix}} = 4\pi/N_{\text{pix}}$); redshift is carried on a grid of $N_{\text{grid}} = 1086$ nodes; a voxel is $v = (p, z)$; coarse radial shells used by the field fit are $g \in \{1, \dots, G_s\}$; catalogs (tracers) are $k \in \{1, \dots, K\}$. Cosmological and population parameters are Λ (containing H_0); survey parameters are $\theta = (\log_{10} n_0, \delta, \theta_{\text{sel}}, b_{\text{miss}}, \dots)$. The expected all-galaxy comoving density is

$$\bar{n}(z; \Lambda, \theta) = \frac{dN_{\text{exp}}}{dz d\Omega} = n_0 (1+z)^\delta \frac{dV_c}{dz d\Omega}(z; \Lambda), \quad (1)$$

and $C(z; \theta) \in [0, 1]$ is the completeness. The production configuration uses the parametric $C_{\text{sel}}(z; \theta_{\text{sel}})$ built from a truncated luminosity function; $f_p = 1 - \text{masked_frac}_p \in [0, 1]$ is the per-pixel areal completeness, which is $f_p \equiv 1$ in today’s code.

The production run this work targets carries 259 events at 4096 posterior samples, 1 067 946 detected injections, 20 free dimensions, and *no large-scale structure at all*: it runs with the field switched off and no completion table. The field-level path would be the first structure ever used in production, which is why the standard is “provably inert when off, provably correct when on.”

1.2 Thinning: why the catalog and the missing galaxies separate

Conditional on a latent log-density contrast $s(p, z)$, galaxies are an inhomogeneous Poisson process with the lognormal intensity

$$\Lambda_{\text{gal}}(p, z) = \bar{n}(z; \Lambda, \theta) \exp\left[b_{\text{gal}} s(p, z) - \frac{1}{2} b_{\text{gal}}^2 \sigma^2(p, z)\right], \quad \sigma^2(p, z) \equiv \text{Var}[s(p, z)], \quad (2)$$

where the shift $-\frac{1}{2} b^2 \sigma^2$ enforces $\mathbb{E}_{\text{prior}}[e^{bs - \frac{1}{2} b^2 \sigma^2}] = 1$: the field *redistributes* galaxies, it does not create them. This mean-one convention is the most load-bearing convention in the stack and it reappears three times below.

The survey observes each galaxy independently with probability $f_p C(z; \theta)$. By the Poisson thinning theorem the catalogued and missing galaxies are two *independent* Poisson processes,

$$\Lambda_{\text{cat}}(p, z) = f_p C(z; \theta) \Lambda_{\text{gal}}(p, z), \quad (3)$$

$$\Lambda_{\text{miss}}(p, z) = [1 - f_p C(z; \theta)] \Lambda_{\text{gal}}(p, z), \quad (4)$$

and, exactly,

$$p(\text{missing} \mid \text{catalogued}, s) = p(\text{missing} \mid s). \quad (5)$$

Equation (5) says: once the field is fixed, the catalog carries no further information about what is missing. Everything the catalog knows about the missing galaxies, it knows through s . It is the entire resolution of the double-conditioning question of §2.4.

1.3 The redshift–sky prior as it is coded

The likelihood does not integrate (3); it uses the catalog itself. For a dark siren at (z, p) ,

$$p(z, p \mid \Lambda, \theta, Q) = \frac{N_{\text{obs}, p} \mathcal{K}_p(z) + [1 - C(z; \theta)] \bar{n}(z; \Lambda, \theta) Q_p(z)}{\mathcal{Z}(\Lambda, \theta, Q)}, \quad (6)$$

[CODE] with $N_{\text{obs},p}$ the integer galaxy count in pixel p and $\mathcal{K}_p$ the pixel’s photometric-redshift kernel density. The consumed missing density is $dN_{\text{miss}} = [1 - C] \bar{n} Q$, relaxed to dN_{exp} above the survey depth z_{depth} — i.e. $C := 0$, $Q := 1$ there. Comparing with (4) identifies

$$Q_p(z) \equiv \exp\left[b_{\text{miss}} s(p, z) - \frac{1}{2} b_{\text{miss}}^2 \sigma^2(p, z)\right] \quad \text{with } f_p \equiv 1. \quad (7)$$

So Q is the exponentiated latent field, mean-one normalised, and nothing else. Its only defect as written is that it is a fixed table read from disk rather than a function of the sampled parameters.

Two structural consequences of $f_p \equiv 1$ deserve naming, because both are first-order for the target survey. The measured areal masking has mean 0.137, standard deviation 0.104 and 99th percentile 0.635 over the occupied footprint, while the magnitude depth is uniform at the retention cut (m_{th} percentiles 1, 50, 99 all equal 21.000; the imaging depth is nowhere the limit). So the entire per-pixel selection variation is *areal*, which is achromatic in redshift and magnitude and therefore exactly correctable. First, putting f_p on the count side only would invert the sign of the inferred incompleteness: a pixel with 63% areal masking would be told that 63% of its galaxies are a *density deficit* rather than a *selection deficit*, and that deficit re-enters the event likelihood as a claim about where hosts are not. Second, the dominant term is not the partial pixels: it is the 18 682 of 49 152 pixels with $f_p = 0$, whose missing-budget weight moves from $1 - C$ to 1, because a pixel with no catalog has *all* of its galaxies missing.

1.4 The offline field: Poisson–lognormal on a low-rank basis

Let $\{Z_j\}_{j=1}^M$ be inducing nodes on $\mathbb{S}^2 \times \zeta$, $\zeta = \log(1+z)$ — a Fibonacci sphere of M_{sph} points crossed with M_z nodes uniform in ζ — and $k = k_{\text{sph}} \otimes k_\zeta$ a separable kernel. With $LL^\top = K_{ZZ} + j\mathbb{K}$, the Nyström design and the whitened field are

$$\Phi = K_{XZ} L^{-\top} \in \mathbb{R}^{V \times M}, \quad s = \Phi \xi, \quad \xi \sim \mathcal{N}(0, \mathbb{K}_M), \quad (8)$$

so $\text{Cov}(s) = \Phi \Phi^\top$ and $\sigma_{\text{prior}}^2(v) = \sum_i \Phi_{vi}^2$. Φ factorises exactly as $\Phi_{\text{sph}} \otimes \Phi_\zeta$ and is never materialised: a dense Φ at $M = 3780$ would be 47.6 GB, and a build at $M = 1728$ has already exhausted memory at 21.7 GB.

The rank is not free. The angular and radial lengthscales must be equal in comoving units or the kernel is a pancake, and the radial scale must survive the photometric-redshift scatter. The measured scatter is $\sigma_z = 0.0227$ (median over 2.24M galaxies), i.e. $\sigma_\chi = 89$ Mpc at $z_{\text{ref}} = 0.237$, so a 50 Mpc radial kernel retains $\exp[-(89/50)^2/2] = 0.20$ of its own radial signal while a 190 Mpc kernel retains 0.90. The self-consistent configuration is isotropic at ~ 190 Mpc: $\ell_{\text{ang}} = 0.2$, $\ell_\zeta = 0.039$, $M_{\text{sph}} = 315$, $M_z = 8\text{--}12$, $M = 2520\text{--}3780$. At that rank roughly 40% of the coefficients are data-constrained and the remainder are prior — which is correct, and is what the Laplace covariance encodes.

The fit is over solve voxels $v = (p, g)$ with integer counts N_v and a base $\bar{\lambda}_v = \int_g C \bar{n} dz$, giving

$$J(\xi) = \frac{1}{2} \|\xi\|^2 + \sum_v [\lambda_v - N_v \log \lambda_v], \quad \lambda_v = \bar{\lambda}_v \exp\left[b(\Phi \xi)_v - \frac{1}{2} b^2 \sigma_{\text{prior}}^2(v)\right], \quad (9)$$

[CODE] with $\nabla J = \xi + b\Phi^\top(\lambda - N)$ and $H = \mathbb{K} + b^2\Phi^\top \text{diag}(\lambda)\Phi \succeq \mathbb{K}$. The problem is convex and well posed even when $V < M$. The Laplace posterior $\mathcal{N}(\hat{\xi}, H^{-1})$ is sampled as

$$\xi_m = \hat{\xi} + L_H^{-\top} g_m, \quad g_m \sim \mathcal{N}(0, \mathbb{K}_M), \quad L_H L_H^\top = H. \quad (10)$$

Lemma 1 (posterior-mean Q and its variance shift). *Under (10), for the Nyström row ϕ_u at an output voxel u ,*

$$\log \mathbb{E}[Q_u] = b \phi_u^\top \hat{\xi} - \frac{1}{2} b^2 [\sigma_{\text{prior}}^2(u) - \sigma_{\text{post}}^2(u)], \quad \sigma_{\text{post}}^2(u) = \|L_H^{-1} \phi_u\|^2. \quad (11)$$

Equation (11) is the table the likelihood consumes today. It has a physically desirable property that a MAP table does not: a voxel with no data has $\sigma_{\text{post}} \rightarrow \sigma_{\text{prior}}$ and $\phi_u^\top \hat{\xi} \rightarrow 0$, so $Q \rightarrow 1$. *No information, no modulation.*

Remark 1 (the plug-in gap, stated exactly). The event likelihood is a nonlinear functional of Q : Q appears inside a logarithm in (6) and inside $\log \mu$ in the detection normalisation. Hence $\log \mathbb{E}_\xi[\mathcal{L}(Q(\xi))] \neq \log \mathcal{L}(\mathbb{E}_\xi[Q])$, and running the deterministic table is a plug-in, exact only to first order in the posterior spread of s .

1.5 The mean-one budget constraint is a gauge fixing

Proposition 1 (Q has no monopole freedom). *For any positive $\alpha(z)$, the map $Q_p(z) \mapsto \alpha(z)Q_p(z)$ together with $[1 - C]\bar{n} \mapsto [1 - C]\bar{n}/\alpha(z)$ leaves dN_{miss} , and therefore the entire event likelihood, exactly invariant. The per- z monopole of Q is therefore not identified by the gravitational-wave data; it is identified only by whatever fixes $(n_0, \delta, \theta_{\text{sel}})$.*

The code fixes this gauge at build time by rescaling each realisation so that

$$\frac{\sum_{p \in \mathcal{F}} w_p(z) Q_p(z)}{\sum_{p \in \mathcal{F}} w_p(z)} = 1, \quad w_p(z) = [1 - C_p(z)] \bar{n}(z), \quad (12)$$

over the *fitted footprint* $\mathcal{F}$, applied to the deterministic table and to *each member independently*. This is necessary and it is measured: the Jensen contribution of Lemma 1 otherwise inflates the missing budget by +55%, because the posterior variance is largest exactly where the data are sparse.

Two precisions matter for what may be claimed. First, in the production completeness modes the constraint is *already* exactly the identity the likelihood consumes: the builder fits the whole sky with p -independent weights, so mean-one reduces to $\langle Q \rangle_{\text{sky}} = 1$ and the consumed sum equals its $Q \equiv 1$ value for every z , member and θ . Under *stratified* selection the weights are p -dependent and evaluated at the build-time fit, so that identity holds only at $\theta = \theta_{\text{ref}}$ and leaks elsewhere at the order of the stratum-to-stratum spread in C_{sel} . Second, and more importantly for the paper: by Proposition 1 this is a *convention that makes an estimand well posed*, i.e. a definition, not a measurement. “ C and n_0 own the budget, Q owns placement” is the correct sentence, and “gauge fixing” is the correct word for it.

Remark 2 (renormalisation does not commute with the ensemble mean). Writing Π for the renormalisation, Π is nonlinear in Q , so $\frac{1}{M} \sum_m \Pi[Q^{(m)}] \neq \Pi[\mathbb{E}[Q]]$ at order $\text{Var}_{\text{sky}}(Q)/N_{\text{pix}}$ per shell. “Members carry placement uncertainty but zero budget uncertainty” is therefore a modelling choice, not a theorem.

1.6 Member marginalisation, and the price of a logarithm

With M_{draw} realisations the likelihood is

$$\log \mathcal{L}(\Lambda, \theta) = \text{LSE} \log \mathcal{L}_m - \log M_{\text{draw}}, \quad (13)$$

[CODE] where the member index enters *both* the per-event terms and the detection normalisation μ — enforced as an all-or-none guard, and correct: a member is one realisation of the universe, and the same universe must place the hosts and set the detected fraction.

Proposition 2 (Jensen bias of (13)). (13) is unbiased for $\mathcal{L}$ and biased for $\log \mathcal{L}$. If $\log \mathcal{L}_m$ has spread σ nats and is approximately Gaussian across members, then

$$\mathbb{E}[\log \hat{\mathcal{L}}_M] - \log \mathcal{L} \simeq -\frac{e^{\sigma^2} - 1}{2M}, \quad \frac{\text{ESS}}{M} \simeq e^{-\sigma^2}, \quad \implies \quad \boxed{M > \frac{e^{\sigma^2} - 1}{2\epsilon}} \quad (14)$$

for an ϵ -nat bias.

This is sharper than the usual $M \sim e^{\sigma^2}$ folklore and it is brutal. At $\sigma = 1.0$ nats, $\epsilon = 0.1$ needs $M = 9$; at $\sigma = 1.5$, $M = 43$; at $\sigma = 2.6$, $M = 4.3 \times 10^3$, and $M = 8$ carries a 54-nat bias. The member spread is therefore the largest open number in the design, and §2.10 predicts it in closed form rather than discovering it.

Remark 3 (what actually has to be gated). The whitened draws g_m are generated once and reused at every θ (common random numbers), so $\hat{\mathcal{L}}_M(\theta)$ is a *deterministic, smooth* function of θ , not a noisy one. The bias of Proposition 2 is a fixed perturbation and only its θ -variation, $\text{osc}_\theta \Delta \equiv \max_\theta \Delta - \min_\theta \Delta$ with $\Delta(\theta) = \log \hat{\mathcal{L}}_M - \log \mathcal{L}$, distorts the posterior; a constant offset is absorbed into the evidence. The consequence is unforgiving: every determinism diagnostic — repeat-to-repeat identity, adjacent- θ smoothness — passes *by construction*, and passes just as well on a badly distorted surrogate. Only $\text{osc}_\theta \Delta$ against a large- M reference discriminates.

1.7 The shared-latent multitracer solve, and which half of the seam is open

For K catalogs sharing one field the stacked objective and Hessian are

$$J(\xi) = \frac{1}{2} \|\xi\|^2 + \sum_k \sum_{v \in \mathcal{V}_k} [\lambda_{kv} - N_{kv} \log \lambda_{kv}], \quad H = \mathbb{K} + \sum_k b_k^2 \Phi_k^\top \text{diag}(\lambda_k) \Phi_k, \quad (15)$$

with $\lambda_{kv} = \bar{\lambda}_{kv} e^{b_k(\Phi_k \xi)_v - \frac{1}{2} b_k^2 \sigma_{kv}^2}$, one Laplace ensemble, and per-survey tables $\log Q_k^{(m)} = b_k \phi^\top \xi_m - \frac{1}{2} b_k^2 \sigma_{\text{prior}}^2$. Member m of every catalog is then the same physical realisation — a theorem, not a check.

The interesting statement is where this stands operationally, and it is not where one would guess. **The consumer is ready; the producer cannot produce.** On the inference side, $K \geq 2$ mixtures under parametric selection completeness assemble and evaluate end to end: per-catalog completeness modes, per-catalog selection fits with an all-or-nothing anchoring rule, a homogeneous luminosity-function family, per-catalog parameter routing, and a two-catalog end-to-end likelihood fixture pinning the identical-catalogs-collapse-to- $K=1$ identity. Exactly one refusal survives on that side — *stratified* fits at $K \geq 2$ — and its stated reason is data plumbing, not modelling.

On the builder side the shared-field path is closed in four independent ways. It takes the per-pixel completeness default and is hard-rejected at load by any parametric-selection run; it never fixes the budget gauge, so every joint file ships the full +55% Jensen monopole that the single-catalog tables remove; its inducing grid is hard-coded at $M_{\text{sph}}, M_z = 32, 6$ up to $z = 3$, giving a ζ node spacing of 0.277 and hence a *hard* floor of $L_{\text{smooth}} \geq 1.34$ Gpc — every physically supportable correlation length fails it; and it is the only producer of a shared realisation identifier, so K independently built single-catalog tables carry K different ones. The only way to run the configuration is a flag whose own documentation states that it marginalises over an *independent-fields product prior* rather than the shared-field prior the estimator assumes.

A shared-latent multitracer marginalisation under parametric selection completeness is not refused. It is not constructible.

This is the cleanest one-line motivation for the whole proposal: the field-level hierarchy closes that seam *by deleting the producer*. One ξ shared across K tracers makes matched realisations a theorem, retires the realisation identifier and the provenance firewall together, and removes the builder’s frozen rank, frozen cosmology and frozen completeness mode simultaneously.

1.8 Novelty, scored honestly

The scoring below is the owner’s, transcribed; the right-hand column is the correction this document’s verification requires.

claim	score	reality check
homogeneous missing galaxies	none	—
$1 + b\delta_g$ correction	low	—
a multiplicative structure field Q	low	variance-completion work introduced it
clustering to reconstruct missing galaxies	not novel	—
Poisson/lognormal density reconstruction	not novel broadly	—
3-D Bayesian dark-siren catalog reconstruction	not novel broadly	cartography-style reconstructions already propagate voxel-count uncertainty and marginalise cosmology and bias; on the θ -coupling axis they are <i>ahead</i> of an exported table
low-rank sphere $\times z$ GP implementation	interesting implementation	—
posterior-mean rather than MAP Q	strong technical choice	correct, and it is what makes $Q \rightarrow 1$ where there is no data
per- z mean-one budget constraint	interesting methodological contribution	it is a gauge fixing (Prop. 1) — a definition, not a measurement; demote to an enabling clause
empty pixels as Poisson information	important, not fundamentally new	—
parametric selection C separated from Q	quite interesting combination	correct, and it is what makes the count likelihood non-circular
marginalising members through a complete GW hierarchy	potentially novel	the same-realisation requirement is a <i>correctness</i> condition, already enforced; the contribution is doing it at fixed budget monopole
joint multitracer shared latent	potentially novel	very not constructible today (§1.7)
joint multitracer + selection C + GW marginalisation	not implemented; strongest direction	unchanged — and the field-level hierarchy is the shortest route to it

Table 1: Novelty scoring, with the corrections the code-level verification forces. The sentence “*we introduce Bayesian large-scale-structure reconstruction for incomplete dark-siren catalogs*” must not be written.

Two scope sentences must ride with any claim to marginalise over the clustered missing field: the marginalisation is over *the volume the survey sees*, not over the budget ($Q \equiv 1$ off-footprint and

above z_{depth} , covering 38% of the sky and $\sim 99.99\%$ of the missing budget); and the compression that makes a defensible M_{draw} affordable is engineering, not a methods claim.

2 Proposed work

2.1 The hierarchy

Promote ξ from a table to a latent variable:

$$\begin{aligned}
p(H_0, \Lambda, \xi, \{\theta_k\}, \{b_k\}, b_{\text{GW}}, \{f_k\} \mid d_{\text{GW}}, \{d_{\text{gal},k}\}) &\propto \pi(H_0, \Lambda) \pi(\{\theta_k\}) \pi(\{b_k\}) \pi(b_{\text{GW}}) \\
&\times \underbrace{\mathcal{N}(\xi; 0, \mathbb{W}_M)}_{\text{(F1)}} \times \prod_k \underbrace{p(\{m_i\}_k \mid \{z_i\}_k, \theta_k)}_{\text{(F2) magnitudes}} \times \prod_k \underbrace{p(\{N_{kv}\} \mid \xi, \theta_k, b_k, f_k, \Lambda)}_{\text{(F3) counts}} \\
&\times \underbrace{p(d_{\text{GW}} \mid \{d_{\text{gal},k}\}, \xi, b_{\text{GW}}, \Lambda, \{\theta_k\})}_{\text{(F4) GW}} \quad (16)
\end{aligned}$$

The single structural change is that ξ now appears in (F1), (F3) and (F4).

An honest statement about (F2). In the shipped pipeline the magnitude channel is *not* a likelihood factor: the magnitude log-likelihood is defined but referenced nowhere outside its own test, and θ_{sel} enters as an anchored Gaussian prior whose width comes from the offline fit’s covariance ($\sigma(\hat{M}_0) = 1.60 \times 10^{-4}$ mag). The shipped hierarchy is therefore empirical Bayes on θ_{sel} , and (16) is the target, not the current object. This matters exactly once: at the unconditional rung of §2.5 the restored shell totals draw θ_{sel} information from the same magnitudes that produced the prior, and the combination is refused.

2.2 (F3): the count likelihood, and the one approximation that buys three things

On solve voxels $v = (p, g)$,

$$N_{kv} \sim \text{Poisson} \left[\underbrace{f_{k,p} C_k(v; \theta_k) \bar{N}_k(v; \Lambda, \theta_k)}_{\bar{\lambda}_{kv}} \exp(b_k s_v - \tfrac{1}{2} b_k^2 \sigma_v^2) \right], \quad (17)$$

with $\bar{N}_k(v) = \Omega_{\text{pix}} \int_g \bar{n} dz$. Two consequences must be stated because both change the completeness side. $f_{k,p}$ appears here *and* in the consumed missing budget $dN_{\text{miss}} = [1 - f_p C] \bar{n} Q$ — one side only inverts the sign of the inferred incompleteness. And under a per-pixel completeness the base $\bar{\lambda}_{kv}$ is itself a plug-in estimate of the same counts, so (17) would be circular; only bases that are functions of θ rather than of the counts are admissible.

Conditioning on the per-shell totals $T_{kg} = \sum_p N_{kpg}$ factorises the counts exactly,

$$p(\{N_{kpg}\} \mid \xi, \theta) = \underbrace{p(\{T_{kg}\} \mid \xi, \theta)}_{\text{monopole}} \times \underbrace{p(\{N_{kpg}\} \mid \{T_{kg}\}, \xi, \theta)}_{\text{placement}}, \quad (18)$$

and using only the second factor is a deliberate *partial likelihood*: the factorisation is exact, the choice is not free. The discarded monopole is removed to linear order by an explicit projection of the sky-constant coefficients and retains $O(b^2 \text{Var}_{\text{sky}}(s)/2)$ content.

The within-shell linearisation. Write the shell-integrated intensity and its normalised within-shell profile,

$$\Lambda_{pg} = a_g(\theta) f_p \int_g \omega_g(z; \theta) e^{bs_p(z) - \frac{1}{2}b^2\sigma_p^2(z)} dz, \quad \omega_g(z; \theta) = \frac{C\bar{n}}{\int_g C\bar{n} dz}, \quad (19)$$

with a_g p -independent. Expanding the integral, $\log \int_g \omega_g e^{bs_p} dz = b\langle s_p \rangle_g + \frac{1}{2}b^2 \text{Var}_g(s_p) + O(b^3)$, and *keeping the first term* defines the shell-response operator W to act on the radial *basis rows* rather than around the exponential:

$$\tilde{\phi}_\zeta[g, :] = \sum_n W_{gn} \phi_\zeta[n, :], \quad W_{gn} = \left(\int_g \mathcal{K}(z|z_n) dz \right) \omega_g(z_n; \theta) \Delta_n, \quad (20)$$

rows normalised to unity, so that

$$\boxed{\eta_{pg} = b \left(\Phi_{\text{sph}}[p] \otimes \tilde{\phi}_\zeta[g] \right) \cdot \xi, \quad \pi_{pg} = \frac{f_p e^{\eta_{pg}}}{\sum_{p'} f_{p'} e^{\eta_{p'g}}}, \quad \log p_{\text{count}} = \sum_g \sum_p N_{pg} \log \pi_{pg}.} \quad (21)$$

This is the load-bearing modelling step, and it is worth being explicit about what it buys and what it costs. *Buys:* η is exactly linear in ξ , so the Kronecker factorisation $\Phi_g = \Phi_{\text{sph}} \otimes \tilde{\phi}_\zeta[g]$ is exact, the Hessian below is exact, and the sky-constant subspace is an *exact* gauge direction of (21). *Costs:* the neglected term is precisely $\frac{1}{2}b^2[\text{Var}_g(s_p) - \langle \sigma_p^2 \rangle_g]$, a within-shell variance, which is p -dependent (which is why it does not divide out) and which vanishes identically when s is constant across a shell. *And this is the same condition that makes the completeness parameters cancel* (Prop. 3 below): one approximation controls the separability, the exactness of the Hessian, and the θ -cancellation together, and its controlling scale is the shell width against the photometric-redshift-smoothed radial structure. Moving W onto the basis is also what makes it a genuine forward model: the photometric-redshift kernel is convolved into the *model*, not deconvolved from the data, so the counts remain integer, the multinomial remains exact, and the radial attenuation $\sim e^{-k^2\sigma_\chi^2/2}$ does not bias the tracer bias.

Gradient and Hessian. For (21),

$$\nabla J_\theta = \xi - b \sum_g \Phi_g^\top (N_g - T_g \pi_g), \quad (22)$$

$$H(\theta) = \mathbb{K} + b^2 \sum_g T_g \left[\Phi_g^\top \text{diag}(\pi_g) \Phi_g - u_g u_g^\top \right], \quad u_g = \Phi_g^\top \pi_g. \quad (23)$$

The rank-one subtraction is not optional: it is positive semidefinite, so omitting it makes H too large, H^{-1} too small, and the Laplace draws *under*-dispersed — exactly the direction that hides the error of Proposition 2. It is separable and therefore free: $u_g = v_g \otimes \tilde{\phi}_\zeta[g]$ with $v_g = \Phi_{\text{sph}}^\top \pi_g$. Equation (23) is the *Fisher* information of the multinomial, hence $H \succeq \mathbb{K}$ for any link function; using the observed Hessian instead would forfeit that guarantee under a saturating link, so Fisher scoring is normative rather than convenient.

2.3 (F4): the gravitational-wave host intensity on the same field

Hosts are built from the *same* s with their own bias,

$$\Lambda_{\text{host}}(p, z; \Lambda, \theta, \xi) = \psi(z; \Lambda) \bar{n}(z; \Lambda, \theta) \exp \left[b_{\text{GW}} s(p, z) - \frac{1}{2} b_{\text{GW}}^2 \sigma^2(p, z) \right], \quad (24)$$

with ψ the merger-rate weighting. Thinning (24) by the same $f_p C$ gives the catalogued/missing split. The point that a cross-correlation-literate reader checks first is the *catalogued* branch: since hosts trace $e^{b_{\text{GW}} s}$ while galaxies trace $e^{b_{\text{gal}} s}$, the probability that a specific catalogued galaxy j hosts the event carries an excess-bias factor. The correct prior is

$$p(z, p \mid \mathcal{C}, \xi, \Lambda, \theta) = \frac{1}{\mathcal{Z}(\Lambda, \theta, \xi)} \left[\sum_{j \in \mathcal{C}} \delta_{p, p_j} \mathcal{K}_j(z) w_j(\Lambda) e^{(b_{\text{GW}} - b_{\text{gal}})s(x_j) - \frac{1}{2}(b_{\text{GW}}^2 - b_{\text{gal}}^2)\sigma^2(x_j)} \right. \\ \left. + [1 - f_p C(z; \theta)] \psi(z; \Lambda) \bar{n}(z; \Lambda, \theta) e^{b_{\text{GW}} s(p, z) - \frac{1}{2}b_{\text{GW}}^2 \sigma^2(p, z)} \right]. \quad (25)$$

At $b_{\text{GW}} = b_{\text{gal}}$ the spike factor is identically unity and the catalogued branch is field-free; away from it, writing the branch field-free is a modelling error that also makes the complete-catalog limit below true by omission rather than by physics. Setting $b_{\text{GW}} \equiv b_{\text{gal}}$ in a headline analysis is therefore not only a statistics choice; it is the choice that makes Limit I exact.

Structurally (25) is (6) with $Q_p(z)$ replaced by an explicit function of ξ , which is why the change in the evaluation is one contraction: $\log Q \rightarrow b_{\text{GW}} (\text{row_fac}[p] \cdot \tilde{\phi}_\xi[n]) - \rho$.

2.4 Double conditioning: the catalog is used once

Theorem 1 (no double use). *Let $\mathcal{C}$ be the catalogued galaxies and $\mathcal{U}$ the missing ones. Then exactly*

$$p(\mathcal{C}, d_{\text{GW}} \mid \xi, \Lambda, \theta) = \underbrace{p(\mathcal{C} \mid \xi, \theta, \Lambda)}_{(\text{F3})} \times \underbrace{p(d_{\text{GW}} \mid \mathcal{C}, \xi, \Lambda, \theta)}_{(\text{F4})}, \quad (26)$$

and the second factor may be evaluated with (25) without any double use of the catalog, provided the missing intensity is written as a function of ξ alone and never of $\mathcal{C}$.

Proof. (26) is the chain rule. The gravitational-wave factor requires the catalogued hosts — a conditioning event, used as known positions, not re-modelled — and the intensity of $\mathcal{U}$, which by (5) is $[1 - f_p C] \Lambda_{\text{gal}}(\xi)$ and involves $\mathcal{C}$ nowhere. The catalog’s information about $\mathcal{U}$ has been routed entirely through (F3) into the posterior for ξ . $\square$

Remark 4 (the two ways it breaks, both live). (i) *Empirical-Bayes plug-in*. If the missing intensity is evaluated at a point estimate $\hat{\xi}(\mathcal{C})$ — which is exactly what the current table is — then $\mathcal{C}$ appears inside the gravitational-wave factor, and multiplying by (F3) uses the same counts twice. The present code is safe *only* because it never multiplies by (F3). **Adding the count likelihood without promoting ξ to a latent variable is the single most dangerous intermediate state available, and it must never exist.** (ii) *The monopole, twice*. The calibration that pins $(\log_{10} n_0, \delta)$ is a fit to the *same* counts over $z \in [0.05, 0.3]$, delivered to the run as fixed values; that prior is a surrogate for the first factor of (18). Keeping both it *and* an unconditional Poisson count likelihood squares the monopole information. Exactly two combinations are admissible: conditional counts + external calibration prior, or unconditional counts + a genuinely independent prior on (n_0, δ) . The first is recommended; the second carries §2.6’s degeneracy.

2.5 Which parameters the field is allowed to couple to

Proposition 3 (exact cancellation). π_{pg} is exactly independent of θ and Λ if and only if the base is p -separable within the shell — true whenever C is sky-uniform — and the field is shell-constant.

Otherwise the residual is exactly a within-shell covariance,

$$\frac{\partial \log \pi_{pg}}{\partial \theta} = \frac{\int_g \partial_\theta \omega_g e^{b_{sp}} dz}{\int_g \omega_g e^{b_{sp}} dz} - \sum_{p'} \pi_{p'g} \frac{\int_g \partial_\theta \omega_g e^{b_{sp'}} dz}{\int_g \omega_g e^{b_{sp'}} dz}, \quad (27)$$

which vanishes identically when $e^{b_{sp}(z)}$ is constant across the shell.

This is the specification of the coupling set, and it is exact. Since $dV_c/dz = (c/H_0)^3 \times \text{shape}(z; \Omega_m)$ and the completeness curve is built h -free, H_0 cancels from (21) identically while $\Omega_m, w_0, w_a, \delta$ and θ_{sel} do not. Three rungs follow, and they are the design space:

rung	structure	parameters reaching the field	per-proposal cost
0	conditional counts, frozen W	none	~ 0
1	conditional counts, live π_{pg}	$\Omega_m, w_0, w_a, \delta, \theta_{\text{sel}}$ (<i>not</i> H_0)	~ 1 ms
2	unconditional counts	the above, plus H_0 through $n_0(c/H_0)^3$	rung 1 + $\log \det H$

Rung 1 is the target. It satisfies “every cosmological proposal changes the galaxy-field likelihood consistently” for every parameter the galaxy field can honestly constrain, and it is affordable. Rung 0 is its control arm and its fallback: it is exactly rung 1 evaluated at $\Delta\theta = 0$.

2.6 The n_0 – H_0 degeneracy: what rung 2 actually costs

Un-conditioning restores the shell-total factor, and the restored channel is

$$T_g(\theta) \propto n_0 (c/H_0)^3 \int_g C_{\text{sel}}(z; \theta_{\text{sel}}) (1+z)^\delta \text{shape}(z; \Omega_m, w_0, w_a) dz, \quad (28)$$

so the *entire* H_0 content of the count channel is the volume–density amplitude $n_0 H_0^{-3}$, exactly degenerate with n_0 . It is identified only if n_0 carries an H_0 -independent prior — and the shipped calibration does not: it computes n_0 by dividing the summed galaxy weights by a comoving volume integral evaluated at the *fiducial* cosmology. Pinning $\log_{10} n_0$ and then sampling H_0 against an unconditional count likelihood therefore manufactures a standard-density H_0 constraint out of the fiducial cosmology, acting against 22.79M galaxies. The honest sentence, which belongs verbatim in any paper:

The galaxy field constrains the shape parameters and the tracer biases. It constrains H_0 only through the volume–density amplitude, which is degenerate with the mean comoving number density; that degeneracy is why n_0 is a nuisance and why its prior must be H_0 -aware rather than a fiducial plug-in.

2.7 Marginalising the field: the galaxy-side evidence

Proposition 4 (the galaxy-side evidence). *With the whitened prior,*

$$\log \mathcal{L}_{\text{gal}}(\theta) = \log \int d\xi \mathcal{N}(\xi; 0, \mathbb{K}_M) p(\{N\}|\xi, \theta) \simeq -J_\theta(\hat{\xi}_\theta) - \frac{1}{2} \log \det H(\theta), \quad (29)$$

with no residual $(2\pi)^{\pm M/2}$: the prior normalisation and the Gaussian integral cancel exactly. Equivalently, with $\ell(\xi, \theta) = \sum_g \sum_p N_{pg} \log \pi_{pg}$,

$$\log \mathcal{L}_{\text{gal}}(\theta) = \ell(\hat{\xi}_\theta, \theta) - \frac{1}{2} \|\hat{\xi}_\theta\|^2 - \frac{1}{2} \log \det H(\theta). \quad (30)$$

The middle term of (30) is an Occam factor and it is *not* removable by the envelope theorem: stationarity gives $\partial_{\xi} \ell(\hat{\xi}) = \hat{\xi}$, not 0, so the envelope theorem removes $\partial \hat{\xi} / \partial \theta$ from J and not from ℓ . Its size is not academic: with $\sim 10^3$ data-constrained modes, $\|\hat{\xi}\| \sim 32$, and a displacement of the mode of order $\|\Delta \hat{\xi}\|_H \sim 6$ across the prior gives $\hat{\xi}^\top S \Delta \theta$ of order a few nats at random alignment and up to $O(10^2)$ nats worst case, against a target accuracy of 0.1 nat. Under linear response it is also free:

$$\frac{1}{2} \|\hat{\xi}_\theta\|^2 = \frac{1}{2} \|\hat{\xi}_{\text{ref}}\|^2 + (S^\top \hat{\xi}_{\text{ref}}) \cdot \Delta \theta + \frac{1}{2} \Delta \theta^\top (S^\top S) \Delta \theta, \quad (31)$$

one precomputed vector and one small matrix. Likewise $\log \det H$ is not *assumed* constant: its motion enters multiplied by $M/2 \simeq 1.9 \times 10^3$, so a 0.1% per-mode drift is 1.9 nats, and it is measured directly at the same solves that produce S — the Cholesky factor is already formed.

2.8 Linear response: the θ -coupled ensemble

Offline, once. From the implicit function theorem, against the same Cholesky factor,

$$S = \frac{\partial \hat{\xi}}{\partial \theta} = -H^{-1} \frac{\partial \nabla_{\xi} J}{\partial \theta} \Big|_{\hat{\xi}_{\text{ref}}, \theta_{\text{ref}}} \in \mathbb{R}^{M \times n_\theta}, \quad (32)$$

i.e. n_θ extra triangular solves and 121 kB of storage at $M = 3780$, $n_\theta = 4$. At $K \geq 2$ each catalog's own selection parameters and its own bias contribute their own *columns* of S : the multitracers generalisation adds terms, it does not restructure anything.

Per proposal. At fixed H , a mean shift is exact:

$$\xi_m(\theta) = \hat{\xi}_{\text{ref}} + S \Delta \theta + L_H^{-\top} g_m \quad (33)$$

is an *exact* draw from $\mathcal{N}(\hat{\xi}_\theta, H^{-1})$ for every θ , with g_m fixed across proposals. There are no importance weights and no effective-sample-size penalty. This is worth stating sharply because the obvious alternative is a trap:

Proposition 5 (the effective-sample-size law for a reused draw set). *If instead one keeps the draws at $\hat{\xi}_{\text{ref}}$ and re-weights, then for a Gaussian mode shift at fixed H , $\text{Var}[\log w] = \|\hat{\xi}_\theta - \hat{\xi}_{\text{ref}}\|_H^2$ exactly, so*

$$\frac{\text{ESS}(\theta)}{M_{\text{draw}}} \simeq \exp\left(-\|\hat{\xi}_\theta - \hat{\xi}_{\text{ref}}\|_H^2\right). \quad (34)$$

The norm is a *sum over all M modes*. A tolerance written per mode as $\|\Delta \hat{\xi}\|_H / \sqrt{M} < \tau$ implies $\text{ESS}/M \simeq e^{-\tau^2 M}$: at $M = 3780$ and a comfortable-looking $\tau = 0.1$ this is 4×10^{-17} . Reusing a fixed draw set across proposals is inadmissible; the mean-shifted construction (33) is the one to build.

What linear response costs. The correction $S \Delta \theta$ is *member-independent*, and that single fact carries the whole cost model. The member row factors stay static under a 256-fold evaluation concurrency; the only per-proposal additions are one member-independent row-factor contraction (~ 1 ms against a 27.5–49.3 ms baseline, and a 1.46 MB transient that must be reserved rather than assumed) and a rank- n_θ update of two sky moments. It is also *analytically smooth* in θ : no solver, no tolerance, no iteration count, no history — so the discontinuity failure mode of a per-proposal re-solve cannot occur by construction.

Why not simply re-solve. Because of arithmetic, not taste. The separable two-stage contraction costs $2G_s N_{\text{pix}} M_{\text{sph}}^2$ for the diagonal part plus $2M_{\text{sph}}^2 G_s M_z^2$ for the radial contraction, and the Cholesky costs $M^3/3$; at $(M_{\text{sph}}, M_z, G_s) = (315, 12, 32)$ that is tens of milliseconds per Newton step and ~ 13 steps, i.e. ~ 1 s per proposal against a 27.5 ms baseline. A re-solve would also drag the member row factors into the per-evaluation transient, 3.0 GB at $M_{\text{draw}} = 8$ and 24 GB at $M_{\text{draw}} = 64$ at production concurrency. The asymmetry, not elegance, is why Proposition 3 is the load-bearing result.

2.9 The budget normaliser in closed form

The normaliser that enforces the gauge of Proposition 1 against the *consumption* weights is

$$\rho_m(z; \theta, b) = \log \frac{\sum_{p \in \mathcal{F}} [1 - f_p C(z; \theta)] e^{b f_m(p, z)}}{\sum_{p \in \mathcal{F}} [1 - f_p C(z; \theta)]} = \log \frac{A_m(z; b) - c B_m(z; b)}{P_{\mathcal{F}} - c F_{\mathcal{F}}}, \quad c \equiv C(z; \theta), \quad (35)$$

with $A_m = \sum_{p \in \mathcal{F}} e^{b f_m}$, $B_m = \sum_{p \in \mathcal{F}} f_p e^{b f_m}$, $P_{\mathcal{F}} = |\mathcal{F}|$, $F_{\mathcal{F}} = \sum_{p \in \mathcal{F}} f_p$. The sums run over the *fitted footprint*: off the footprint $f_p = 0$, the consumption weight is unity and $Q \equiv 1$, so that block satisfies the budget identity trivially and must not absorb the footprint’s monopole. A_m and B_m are θ -free and are computed once on a grid in b ; the only online θ -dependence is the scalar c at each node, so (35) is exact rather than interpolated. This is what removes a full-sky reduction — measured at 25.6 ms in double precision, i.e. +50–90% of the entire likelihood — from the per-proposal path. Under linear response the moments acquire θ -dependence only through the same projection, $\partial A_m / \partial \theta_j = \sum_p e^{b f_m^{\text{ref}}} (\Phi_p \cdot S_j)$, at 1.1 MB of extra storage.

The identity the design must preserve, exactly, per realisation, is

$$\frac{\sum_{p=1}^{N_{\text{pix}}} [1 - f_p C(z; \theta)] Q_p(z)}{\sum_{p=1}^{N_{\text{pix}}} [1 - f_p C(z; \theta)]} = 1 \quad \text{for every } z, \text{ member and } \theta. \quad (36)$$

2.10 The member estimator, predicted before it is measured

The largest open number in the design is the member spread σ of Proposition 2, because M_{draw} depends on it exponentially. It can be predicted in closed form from objects that already exist. To first order in the field, the gradient of the total event log-likelihood with respect to the latent is

$$a \equiv \nabla_{\xi} \left(\sum_i \ell_i \right) = b_{\text{GW}} \left(\sum_i \varphi_i \Phi_i - N_{\text{obs}} \langle \Phi \rangle_{\text{sel}} \right), \quad \sigma = \|L_H^{-1} a\|_2 \quad (37)$$

where φ_i is event i ’s fraction of prior mass in the missing branch and $\langle \Phi \rangle_{\text{sel}}$ is the selection-weighted basis average. The norm is the H^{-1} norm, not the Euclidean one: for members drawn as in (10), $\text{Var}_m(\ell_m) = a^\top H^{-1} a$ exactly. The subtraction term is the detection normalisation, and it is what makes (37) a correlation statistic rather than a raw pair count.

The same object a does three jobs, which is the reason to build it first. It predicts σ and hence M_{draw} . It gives the closed-form target for the small-bias limit of the estimator: expanding (13) to second order,

$$\text{LSE}_m \ell_m - \log M_{\text{draw}} - \ell(\hat{\xi}) \longrightarrow \frac{1}{2} a^\top H^{-1} a = \frac{1}{2} \sigma^2, \quad (38)$$

so prediction and validation become one measurement, checked against truth rather than against a larger draw count. (With the counts switched off, $\hat{\xi} = 0$ and $H = \mathbb{K}$, and (38) reduces to the pure prior form $\frac{1}{2} b_{\text{GW}}^2 \sum_{ij} K(x_i, x_j)$ — the two-arm structure is what makes this a genuine closed-form test rather than a certificate of prior collapse.) And it supplies the criterion of the next subsection.

2.11 Two numbers decide the whole design

What decides whether the field must be coupled to the proposal is not whether the fitted field *looks* different — it is whether the *likelihood* is different, in nats. To first order $\Delta \log \mathcal{L} = a \cdot \Delta \hat{\xi}$, and Cauchy–Schwarz gives $|\Delta \log \mathcal{L}| \leq \sigma \|\Delta \hat{\xi}\|_H$. A per-mode tolerance is therefore indeterminate by $\sqrt{M} \simeq 61.5$ depending on an alignment between the field displacement (a radial re-weighting of shells) and the event gradient (the localisation volumes) that nothing bounds and that has every reason to be non-negligible, since both live in the same radial direction. Do not bound the alignment; measure it. The two criteria, both in nats, both computable from the same set of solves at prior-drawn θ :

$$\text{osc}_{\theta} \Delta \log \mathcal{L}_{\text{GW}} = \text{osc}_{\theta} \left[a \cdot (\hat{\xi}_{\theta} - \hat{\xi}_{\text{ref}}) \right], \quad (39)$$

$$\text{osc}_{\theta} \Delta \log \mathcal{L}_{\text{gal}} = \text{osc}_{\theta} \left[\ell(\hat{\xi}_{\theta}, \theta) - \frac{1}{2} \|\hat{\xi}_{\theta}\|^2 - \frac{1}{2} \log \det H(\theta) \right]. \quad (40)$$

Remark 5 (they are not the same number). It is tempting to argue that if the field barely moves then the galaxy-side evidence cannot be informative either. That is false, and the direction of the error matters. The evidence responds to the *full* count residual $r = N - T\pi$ contracted with $\partial_{\theta}\eta$; the field displacement is that residual *projected onto* $\text{span}(\Phi)$ and damped by H^{-1} . A θ -direction lying inside $\text{span}(\Phi)$ is absorbed by the field — large displacement, flat evidence, no constraint. A direction orthogonal to it cannot be absorbed — zero displacement, maximal evidence constraint. Since (27) makes the θ -direction exactly a *within-shell* covariance, and a shell-collapsed basis cannot represent within-shell structure, “small displacement with a live galaxy-side constraint” is the structurally expected case, not an excluded one. (39) and (40) must both be measured and reported.

Remark 6 (and the galaxy-side channel must not be quoted as cosmology). $\Delta \log \mathcal{L}_{\text{gal}}$ acts against 22.79M galaxies while the gravitational-wave side carries 259 events, so it can dominate the posterior. But it is a dN/dz -shape constraint, exactly degenerate with the radial evolution index δ , with the population-average photometric-redshift kernel and with the shell binning — all of which are frozen inside W and none of which are sampled. It is therefore reported as a diagnostic and does not enter a headline posterior unless W ’s own parameters are sampled or profiled. Otherwise the headline result is a galaxy-clustering measurement wearing a dark-siren hat.

2.12 Three limits, one hierarchy

Write the single-event likelihood as

$$\mathcal{L}_i(\Lambda, \theta, \xi) = \int dz d\hat{n} p_i(d_i | d_L(z; \Lambda), \hat{n}) \frac{\Lambda_{\text{cat}}(\hat{n}, z) + \Lambda_{\text{miss}}(\hat{n}, z; \xi)}{\int [\Lambda_{\text{cat}} + \Lambda_{\text{miss}}]}. \quad (41)$$

All three established analyses are limits of (41), and marginalising the field is exactly the operation that interpolates between them.

Limit I — complete catalog, good localisation. $f_p C \rightarrow 1$ gives $\Lambda_{\text{miss}} \rightarrow 0$, and at $b_{\text{GW}} = b_{\text{gal}}$ the catalogued branch of (25) is field-free, so

$$\mathcal{L}_i(\Lambda) \longrightarrow \sum_{j \in \Omega_{90}^{(i)}} w_j(\Lambda) p_i(d_i | d_L(z_j; \Lambda), \hat{n}_j), \quad (42)$$

the classical weighted-host dark siren, with ξ dropping out of (41) *identically*, member by member. **Exact.** This is a physics identity, not a routing property, and it is the strongest inertness statement

the design admits — but only at $b_{\text{GW}} = b_{\text{gal}}$; otherwise the excess-bias factor of (25) survives and the limit is deliberately not an identity.

Limit II — incomplete, good localisation. Both terms live. Because a well-localised event spans $O(1\text{--}10)$ correlation cells at the smoothing scale, the likelihood depends on the field essentially through its point value in the localisation volume, and

$$\log \mathbb{E}_\xi[\mathcal{L}_i] \simeq \log \bar{\mathcal{L}}_i + b_{\text{GW}} \langle s \rangle_i + \frac{1}{2} b_{\text{GW}}^2 \left[\text{Var}_\xi \langle s \rangle_i - \langle \sigma^2 \rangle_i \right] + O(b^3). \quad (43)$$

The first-order term is the placement *information*; the second-order term is the placement *uncertainty*, and it is exactly what the member ensemble estimates. This is the regime every real catalog/detector pairing occupies.

Limit III — diffuse localisation: a correlation-function statement. Let the catalogued fraction of each event’s prior mass tend to zero while the localisation spans many correlation cells. The galaxy counts are *retained* — they are what carries b_{gal} . Linearising both the event factor and the count factor in the field and marginalising the Gaussian latent jointly,

$$\log \int d\xi \mathcal{N}(\xi; 0, \mathbb{K}) e^{(a_{\text{GW}} + a_{\text{gal}}) \cdot \xi} = \frac{1}{2} \|a_{\text{GW}}\|^2 + a_{\text{GW}} \cdot a_{\text{gal}} + \frac{1}{2} \|a_{\text{gal}}\|^2, \quad (44)$$

with $a_{\text{GW}} = b_{\text{GW}} \sum_i \Phi_i$ and $a_{\text{gal}} = b_{\text{gal}} \sum_{p,g} (N_{pg} - T_g \pi_{pg}) \Phi_{pg}$. *Marginalising the Gaussian latent replaces the field by its kernel*, and every pair of field-coupled objects contributes $b_a b_b K(x_a, x_b)$: event $\times$ event is the gravitational-wave auto-correlation; event $\times$ galaxy-voxel is

$$b_{\text{GW}} b_{\text{gal}} \sum_i \sum_{p,g} (N_{pg} - T_g \pi_{pg}) K(x_i, x_{pg}) \longrightarrow b_{\text{GW}} b_{\text{gal}} \sum_\ell \frac{2\ell + 1}{4\pi} \mathcal{W}_\ell^i \mathcal{W}_\ell^g C_\ell^{\text{GW} \times g}, \quad (45)$$

the harmonic cross-correlation estimator, with the per-event localisation windows $\mathcal{W}_\ell^i$ playing the role of tomographic bin windows and the bias combination $b_{\text{GW}} b_{\text{gal}}$ being exactly what such a measurement constrains; and voxel $\times$ voxel is the galaxy auto-correlation, i.e. the count likelihood itself. **Approximate:** (44)–(45) are the $O(b^2)$ truncation of (41), valid in the Gaussian, weakly clustered regime, and the expansion parameter b_{GW} over the events’ support is a measured quantity, not an assumed one.

The unification statement, and its honest scope. Limits I–III are not three methods; they are one likelihood at three completeness/localisation regimes, and the field-level hierarchy is the interpolant that is valid at both ends and, unlike either, in the middle. Two scope statements ride with this. Limit III is a property of *the hierarchy*; the practical estimator conditions on the galaxy data first, absorbing the counts into $\hat{\xi}$ and H , so the cross term is not a separate object of the estimator. And at rung 0 that conditioning happens at frozen θ , which removes precisely the cosmology dependence of $C_\ell^{\text{GW} \times g}$ that a cross-correlation measurement uses — which is an argument *for* the θ -coupled rung, not against the unification.

2.13 The algorithm

Algorithm 1 Offline anchor construction (once per catalog set)

Require: catalogs $\{k\}$, areal-completeness maps $\{f_k\}$, anchor $(\theta_{\text{ref}}, \Lambda_{\text{ref}})$, bias grid $\{b\}$

- 1: **basis:** inducing nodes $M_{\text{sph}} \times M_z$; $L \leftarrow \text{chol}(K_{ZZ} + j\mathbb{K})$; form Φ_{sph} and Φ_ζ {never a dense Φ : 47.6 GB at $M = 3780$ }
 - 2: **counts:** $N_{kpg} \leftarrow \text{histogram}(z_{\text{gal}}; \text{shell edges})$ {integer, exact; edges stamped, never inherited from a global z_{max} }
 - 3: **shell response:** W_{gn} by (20), rows normalised; $\tilde{\phi}_\zeta \leftarrow W\phi_\zeta$ {photo- z convolved *forward*, into the basis}
 - 4: **solve:** $\hat{\xi}, H \leftarrow$ Fisher scoring on $J_{\theta_{\text{ref}}}$ with (22)–(23) {separable two-stage contraction}
 - 5: **sensitivities:** $S \leftarrow -H^{-1}\partial_\theta \nabla_\xi J$ by (32); also $\partial\hat{\xi}/\partial b$ $\{n_\theta+1$ triangular solves against the same L_H ; 121 kB}
 - 6: **Occam pre-contraction:** store $S^\top \hat{\xi}_{\text{ref}}$ and $S^\top S$ {makes (30) free online}
 - 7: **draws:** $\xi_m = \hat{\xi} + L_H^{-\top} g_m$, g_m in antithetic pairs, seed stamped
 - 8: **moments:** A_m, B_m on the b -grid over $\mathcal{F}$, and their projections $\partial A_m / \partial \theta$, $\partial B_m / \partial \theta$
 - 9: **stamp:** content hash over basis, kernel, jitter, N_{side} , f_p , W , shell edges, counts, θ_{ref} , $\{b_k\}$
-

Algorithm 2 Per-proposal likelihood, rung 1 (linear response)

Require: $(H_0, \Lambda, \{\theta_k\}, \{b_k\}, b_{\text{GW}})$; anchor artefact of Alg. 1

- 1: $\Delta\theta \leftarrow \theta - \theta_{\text{ref}}$ $\{H_0$ is absent by Prop. 3}
 - 2: $\Delta\Xi \leftarrow \text{reshape}(S\Delta\theta, (M_{\text{sph}}, M_z))$ {member-independent: this is the whole cost model}
 - 3: $\log \mathcal{L}_{\text{gal}} \leftarrow \ell_{\text{ref}} + \partial_\theta \ell \cdot \Delta\theta - \frac{1}{2} \|\hat{\xi}_{\text{ref}}\|^2 - (S^\top \hat{\xi}_{\text{ref}}) \cdot \Delta\theta - \frac{1}{2} \Delta\theta^\top (S^\top S) \Delta\theta - \frac{1}{2} \log \det H$
 - 4: $c(z) \leftarrow C(z; \theta)$ {a scalar per redshift node}
 - 5: $A \leftarrow A^{\text{ref}} + (\partial A / \partial \theta) \Delta\theta$; likewise B ; $\rho_m \leftarrow \log[(A_m - cB_m) / (P_{\mathcal{F}} - cF_{\mathcal{F}})]$ by (35)
 - 6: **for** each member m **do**
 - 7: $\log Q_m(p, z) \leftarrow b_{\text{GW}}[(\text{row_fac}_m[p] + \Delta\Xi[p]) \cdot \tilde{\phi}_\zeta[z]] - \rho_m(z)$ {0 exactly for $p \notin \mathcal{F}$ and for $z > z_{\text{depth}}$ }
 - 8: $\ell_m \leftarrow \text{EVENTTERMS}(\log Q_m) - N_{\text{ev}} \log \mu(\log Q_m)$ {same realisation in both, enforced}
 - 9: **end for**
 - 10: $\log \mathcal{L} \leftarrow \log \mathcal{L}_{\text{gal}} + \text{LSE}_m \ell_m - \log M_{\text{draw}}$
 - 11: report member ESS = $\exp(-\sum_m p_m \log p_m)$, $p_m = \text{softmax}_m(\ell_m)$, and the two channels of (39)–(40) separately
-

2.14 What is exact and what is not

3 Implementation

3.1 The ladder

Legacy behaviour is bit-identical throughout, enforced by a golden test at 10^{-12} relative tolerance in exact mode; regenerating that golden is forbidden at every step. Two early exits are built in by design, at day 3 and at day 14.

statement	status	controlling parameter
Thinning independence (5)	exact	—
Chain-rule factorisation (26)	exact	—
Posterior-mean Q , Lemma 1	exact	given the Laplace posterior
Monopole gauge, Prop. 1	exact	—
Shell-total factorisation (18)	exact	(using one factor is a <i>choice</i>)
Kronecker separability and (23) for (21)	exact	given the linearisation below
Sky-constant gauge direction of (21)	exact	given the linearisation below
θ -cancellation, Prop. 3	exact iff	p -separable base <i>and</i> shell-constant s
Mean-shifted draws (33)	exact	at fixed H
Effective-sample-size law (34)	exact	Gaussian mode shift at fixed H
Limit I (42)	exact	$f_p C \rightarrow 1$ and $b_{\text{GW}} = b_{\text{gal}}$
Low-rank $\Phi\Phi^\top \approx K$	approx.	rank vs. $\ell_{\text{ang}}, \ell_\zeta$; hard guard
Within-shell linearisation (21)	approx.	$\frac{1}{2}b^2[\text{Var}_g(s_p) - \langle\sigma_p^2\rangle_g]$
Laplace evidence (29)	approx.	non-Gaussianity of the <i>count</i> term on data-rich voxels
Member estimator (13)	approx.	$(e^{\sigma^2} - 1)/2M$, eq. (14)
Deterministic- Q plug-in, Rem. 1	approx.	first order in posterior spread
Renormalise-then-average, Rem. 2	approx.	$\text{Var}_{\text{sky}}(Q)/N_{\text{pix}}$
Frozen H in the draw covariance	approx.	$\frac{1}{2}\sigma^2\epsilon_H$; 0.1–0.3 nat at 10% drift
Frozen log det H	approx.	$\times M/2 \simeq 1.9 \times 10^3$ per unit relative drift
Linear response $\hat{\xi}_\theta \approx \hat{\xi}_{\text{ref}} + S\Delta\theta$	approx.	second order in $\Delta\theta$
Population-average photo- z kernel in W	approx.	spread of σ_z across the sample
Limit II (43)	approx.	$O(b_{\text{GW}}^3)$
Limit III (45)	approx.	$O(b^2)$ truncation; Gaussianity
$Q \equiv 1$ above z_{depth} and off-footprint	<i>choice</i>	zero assumed variance where there is no data
Empirical-Bayes θ_{sel} prior instead of (F2)	<i>choice</i>	anchored at 1.6×10^{-4} mag

Table 2: The ledger. The “choice” rows are modelling conventions and are quoted as systematics, never as marginalisations.

step	scope	gate	days
0	Can the field move the answer at all? $\sum_i \varphi_i$ and σ from (37); the oscillation of $\log \mathcal{L}$ across the H_0 prior with the <i>existing</i> table switched on at the anchor; a three-column cost baseline (no structure / table / latent); the paper-by-paper novelty search	kill criterion K0 ; all four artefacts committed. Nothing downstream is usable until this lands	3
1	Basis module: Kronecker design, named jitter convention, shell response (20), sky moments, prior variances; builders rewired to call it	Kronecker identity to 10^{-13} against a factored reference at production rank; builder outputs byte-identical	3

step	scope	gate	days
2	Areal completeness as data: f_p at $N_{\text{side}} = 64$; counts as an inference input; $C_p = f_p C(z)$ on <i>both</i> sides behind its own flag and its own golden	counts round-trip exactly; sky area to 1%; flag off is bit-identical; flag on moves the missing budget in the predicted direction and by the predicted magnitude; K8 on a non-railing configuration	5
3	Count channel, θ-shaped from the start: (21), (22)–(23), Fisher-scoring solve, sensitivities (32) built from the stacked K -tracer objective, antithetic draws	separable Hessian vs. dense to 10^{-10} ; solver vs. dense Newton to 10^{-8} ; within-shell Jensen residual reported; both criteria (39)–(40), the log-det drift, and the linear-response residual — kill criterion K9	5
4	Anchor artefact: $\hat{\xi}$, L_H , S , Occam pre-contractions, (A, B) and their projections, members, row factors, stamp	build completes at $M = 3780$ in minutes; moment tables to 10^{-6} ; projected moments to the same scale; draws reproduce the posterior mean; hash reproducible	4
5	The evaluation seam, inert by default: latent leaves built and barriered outside the traced body; closed-form ρ ; memory accounting on both the static and the reserved-transient path	every golden cell bit-identical with the flag off; complete-catalog limit (42) bit-identical per member at $b_{\text{GW}} = b_{\text{gal}}$; off-footprint rows return exactly zero; reservation within 10% of measured peak	4
5b	Member spread: predict σ by (37), then measure it at $M_{\text{draw}} = 256$ across the H_0 prior	predicted vs. measured σ reconciled; closed-form limit (38); θ -varying bias < 0.1 nat at the chosen M_{draw} ; K5	2
6a	The compressed ensemble (control arm, and the fallback): rung 0 live at the anchor; budget identity (36); bias-parameter inversion; provenance replaced by the artefact stamp	budget identity to 10^{-12} for all z , members and θ ; closure tiers A–D; overhead inside the accuracy budget; K2, K4	5
6b	The θ-coupled likelihood — the deliverable: linear response (33), projected moments, the evidence (30), the two-channel report	at $\Delta\theta = 0$, bit-identical to 6a; linear-response residual rechecked at production rank; no adjacent- θ discontinuity over 10^4 points; measured overhead ~ 1 ms	5
6c	<i>Optional:</i> unconditional counts, restoring H_0 through $n_0(c/H_0)^3$; Laplace log-det live; saturating link	agrees with 6b where conditioning is benign; K10 ; gradient-norm stopping rule with the residual inside the oscillation budget	5

step	scope	gate	days
7	Multitracer $K \geq 2$: per-tracer bias, completeness, counts and columns of S ; the realisation identifier retires	$K = 1$ bit-identical; bias ratio recovered within 2σ over 20 realisations; shared-field coupling demonstrably tighter than two independent fits , shown against a deliberately decoupled variant	4
8	Support and amplitude profile above z_{depth}	delivered as a <i>sensitivity scan</i> — H_0 shift versus assumed amplitude — never as a marginalised posterior	3
10	Performance and the production campaign	chunk plans; sample compaction below z_{depth} ; the run	3
11	The paper	the day-3 novelty search is cited; the scope sentences of §1 and the H_0 sentence of §2.6 appear verbatim	3

Critical path: **31 days** to the compressed ensemble, **36** to the θ -coupled likelihood, with exits at day 3 and day 14.

3.2 Cost

The baseline is measured, not modelled: 27.5 ms per likelihood call in a single pass and 49.3 ms under a poorly chosen block plan, at production dimensions, with 256 concurrent evaluations and 10.4 GiB of 72.7 GiB already static. The production configuration carries *no* structure at all, so the honest added cost of the deliverable is the whole member-marginalised slice, not the difference against a table-marginalised run:

M_{draw}	member-dependent slice, table	member-dependent slice, latent	added vs. the 27.5 ms baseline
8	5.3 ms	8.6 ms	+31%
32	21 ms	34.4 ms	+125%
64	42 ms (a 13.7 GB member cube)	68.8 ms (a 94 MB row factor)	+250%

Three structural facts hold this together. The compression from a member *table* cube to member *row factors* is a factor $N_{\text{grid}}/M_z \simeq 90$ (1.71 GB $\rightarrow$ 11.7 MB at $M_{\text{draw}} = 8$), which is what makes a defensible M_{draw} affordable at all under (14). The full-sky budget reduction leaves the per-proposal path entirely, in closed form, by (35). And the θ -correction is member-independent, so the resident latent arrays stay static (+140 to +225 MB) and the only per-evaluation transient is the 1.46 MB row expansion of $S\Delta\theta$, i.e. ~ 375 MB at 256-fold concurrency — reserved explicitly, not assumed. Under any per-proposal re-solve variant every one of those three facts fails simultaneously: the moments regain θ -dependence (+50–90%), the row factors become transient (3–24 GB), and the evaluation goes memory-bound before it goes compute-bound.

3.3 When to stop

K0 (day 3).

If the event-weighted in-support prior mass is below 10^{-3} *and* switching the existing table on at the anchor moves the log-likelihood by less than 0.1 nat across the H_0 prior, the field cannot move the posterior. Stop. The deliverable is the bounded systematic, and it is three days away rather than thirty-six. (Context: the in-support missing budget is $\sim 2 \times 10^{-5}$ of the total by three independent integrals; the production H_0 posterior is already identical with and without a table; and most event posterior samples lie above z_{depth} .)

K2.

If neither latent-on versus latent-off nor θ -coupled versus frozen moves the H_0 median by 0.1σ or the 90% interval width by 5%, on closure *and* on a non-railing production configuration, stop at that rung and publish the bound as a systematic.

K3.

Fitted-versus-truth slope below 0.8 at the guard-compliant rank: the basis is collapsing (the pathological value is 0.04).

K4.

Added wall beyond the accepted budget with the accuracy target unmet: descend one rung at a time — θ -coupled $\rightarrow$ frozen ensemble $\rightarrow$ fixed-realisation systematic — rather than jumping to the weakest fallback.

K5.

Member effective sample size below 2 of M_{draw} with the θ -varying bias unmeetable at $M_{\text{draw}} \leq 128$: do not ship it as a marginalisation.

K8.

The areal-completeness change is *expected* to be large — the 18 682 empty pixels alone move the all-sky missing budget by $\sim +45\%$ — so if it moves the H_0 posterior by more than 1σ with the field off, that change is a result in its own right and ships as its own arm; the field programme continues behind it.

K9 (day 14).

If (39) exceeds 0.1 nat and the linear-response residual cannot be brought below 0.1 nat with a second-order term and multiple anchors, then a re-solve is required and a re-solve is not affordable: ship the frozen ensemble and quote the measured θ -dependence as a systematic. Symmetrically, if (39) is below 0.1 nat the coupling adds nothing on the gravitational-wave side — but by Remark 5 that says nothing about (40), which is gated separately.

K10.

If the unconditional rung is taken and n_0 is not sampled under an H_0 -aware prior (or reparameterised H_0 -invariantly), and likewise for θ_{sel} against its anchored magnitude prior, any H_0 shift is an artefact of the fiducial calibration. Refuse the arm rather than publish it.

3.4 What not to attempt

A full per-proposal Newton re-solve ($\sim 36\times$ the entire likelihood, and it makes the member arrays transient). Reusing one draw set across proposals with importance weights ($\text{ESS}/M = 4 \times 10^{-17}$ at a per-mode displacement that reads as comfortably tight). Adding the count likelihood while leaving the field a plug-in $\hat{\xi}(\mathcal{C})$ — Remark 4(i). Materialising Φ (47.6 GB at $M = 3780$). Sampling

ξ in the outer chain (the production run has 20 free dimensions; $M \geq 2520$ is not a perturbation on that). Widening shells past the photometric-redshift scatter. And building the multitracers table producer: the field-level path makes it unnecessary, and it would inherit that producer’s missing budget gauge and its 1.34 Gpc rank floor. One caution on rank: a feasibility argument made at $M \sim 10^2$ is made at a rank the resolution and isotropy guards forbid — feasibility here comes from the correction being member-independent and of rank n_θ , not from a small M .

4 Decisions required

- D1. Which rung?** (a) conditional counts, frozen response: no θ coupling; the control arm and the fallback. (b) conditional counts, linear response: coupling for $\Omega_m, w_0, w_a, \delta, \theta_{\text{sel}}$, no H_0 — **recommended**, conditional on the day-14 criterion. (c) unconditional counts: adds H_0 only through $n_0(c/H_0)^3$, admissible only with D6 resolved and §2.6 stated in the paper. (d) unconditional counts with a per-proposal re-solve: infeasible. *Trade*: conditioning gives up the dN/dz constraint on the budget, restored explicitly as a prior rather than as a by-product of the field.
- D2. Jitter convention** for the Kronecker basis. *Recommend* an absolute, amplitude-independent 10^{-6} on both factors, stamped. *Cost*: latent-mode Q differs from any legacy table by $\sim 2 \times 10^{-3}$ at the guard rank, so every migration comparison references a same-convention rebuild.
- D3. Smoothing scale.** *Recommend* the isotropic ~ 190 Mpc kernel ($\ell_{\text{ang}} = 0.2$, $\ell_\zeta = 0.039$, $M_{\text{sph}} = 315$, $M_z = 8\text{--}12$). *Cost*: the analysis is explicitly a 190 Mpc-smoothed density field. *Alternative*: keep 50 Mpc radially and accept a 4:1 anisotropic kernel measuring 20% of its own radial signal — which must then be stated as such.
- D4. Per-pixel areal completeness.** *Recommend* f_p on both sides with $C_p = f_p C(z)$ in the missing budget — exact for areal masking. *Fallback*: restrict the count channel to a mask-homogeneous sub-footprint, exact without touching the completeness side, losing $\sim 30\%$ of area. Not an option: ignoring it, which inverts the sign of the inferred incompleteness.
- D5. Marginalisation-accuracy budget, hence M_{draw} .** *Recommend* 0.1 nat of θ -varying bias across the H_0 prior, with M_{draw} set against (14) and σ measured in the H^{-1} norm of (37). *Cost, against the true baseline*: +31% at $M_{\text{draw}} = 8$, +125% at 32, +250% at 64. *Alternative*: the fixed-realisation systematic — zero cost, reduced claim. Note this arithmetic holds only under linear response.
- D6. Budget uncertainty.** *Recommend* sampling $(\log_{10} n_0, \delta)$ under the calibration covariance, systematics-inflated, rather than the current plug-in from the same counts. **Required, not recommended, under D1(c)**: the calibration must be re-derived as a function of H_0 , or n_0 reparameterised to an H_0 -invariant combination such as $n_0 h^{-3}$.
- D7. Above the survey depth.** *Recommend* $Q \equiv 1$ with zero variance in the headline analysis, with the amplitude profile delivered as a sensitivity scan quoted as “ H_0 shift under an assumed amplitude”, never as a marginalised posterior — because no measurement in this programme constrains the field there.
- D8. Is the gravitational-wave bias free, and what does that imply for the catalogued branch?** *Recommend* $b_{\text{GW}} \equiv b_{\text{gal}}$ in the headline run: it makes Limit I exact and the complete-catalog test a genuine physics identity, at zero cost. Wherever b_{GW} is free — the mock

campaign, the multitracer tier — the excess-bias factor $e^{(b_{\text{GW}}-b_{\text{gal}})s(x_j)}$ **must** be written on the catalogued spike weights. Writing that branch field-free is a modelling error, and it is the first thing a cross-correlation-literate referee checks.

- D9. Tracer overlap at $K \geq 2$.** *Recommend* a disjoint partition. The event-side mixture is safe under overlap; the count-side product is not.
- D10. Field family for the headline run, and whether to keep the table path.** The production table is radial, and the radial builder is a per-pixel quasi-Newton loop that cannot move inside the likelihood at any rung — so adopting the latent path *is* a change of family, by necessity rather than preference. *Recommend* running four arms of one comparison before choosing — radial table / low-rank table / frozen latent / θ -coupled latent — and keeping the table path indefinitely as the golden-pinned reference. Note that the $K \geq 2$ parametric-selection table arm cannot be built today at any K (§1.7).
- D11. The novelty search.** Resolved by scheduling: it is the first item of work, not the last. No “first” appears anywhere until it is written down.
- D12. What is the headline?** Strip what is already scored as not novel and the budget-preserving marginalisation reduces to a gauge fixing (Prop. 1) plus a correctness requirement (same realisation in the event terms and the normalisation, already enforced); with the mandatory scope caveats what remains is “*we marginalise over the angular placement of missing hosts inside the surveyed volume at fixed budget.*” *Recommend (a):* headline the **unification** — Limits I–III as one likelihood, with the θ -coupled field as the object that makes the interpolation real — and demote the budget-preserving marginalisation to the enabling clause it is. *(b)* headline the marginalisation and accept that it may read as a methods note. *(c)* defer until the day-3 search returns.
- D13. Does the galaxy-side evidence enter the headline posterior?** It acts against 22.79M galaxies while the events number 259, and it is a dN/dz -shape constraint degenerate with δ , with the photometric-redshift kernel and with the shell binning — all frozen inside W . *Recommend (a):* report it as a diagnostic; it does not enter the headline posterior unless W ’s own parameters are sampled or profiled. *(b)* admit it and sample/profile those parameters — real work, and the only honest route to quoting cosmology from it. *(c)* admit it with W frozen: refused.
- D14. Is the magnitude channel a likelihood factor or an anchored prior?** Today it is an anchored Gaussian prior at 1.6×10^{-4} mag; the magnitude log-likelihood exists but is never evaluated. *Recommend (a):* keep empirical Bayes, state it plainly, and let the rung-dependent guards carry the consequences — zero work, honest. *(b)* implement it as a likelihood factor and drop the anchored prior: this closes (16) exactly and is what would permit the unconditional rung without double-counting θ_{sel} , at the cost of real work.

The two decisions that gate everything else are D1 and D12, and both are answerable within three days of starting: D1 by the day-3 measurement of whether the field can move the posterior at all, and D12 by the paper-by-paper search that the same three days contain.